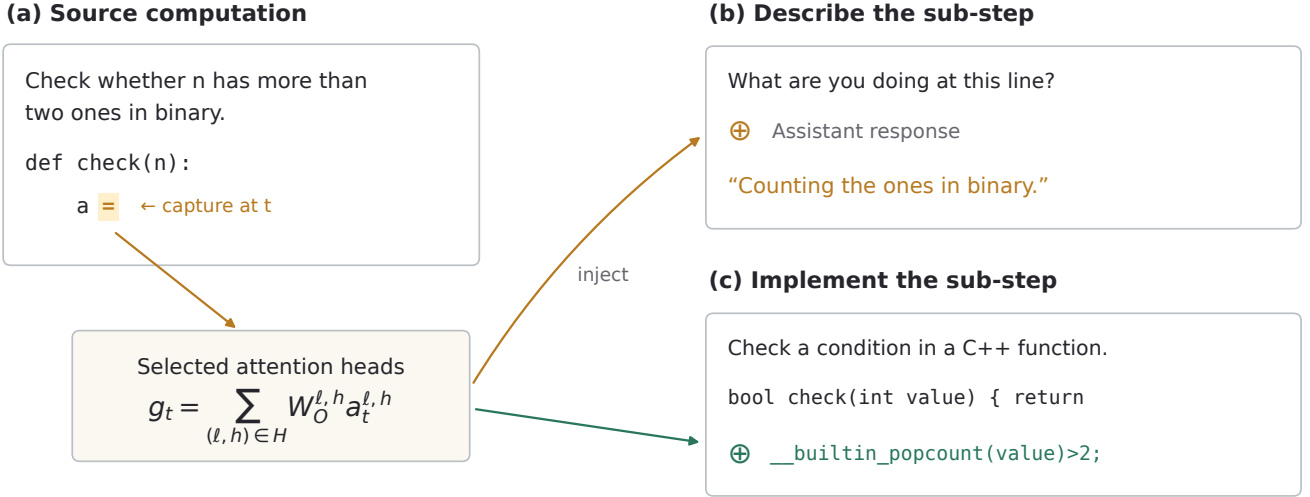
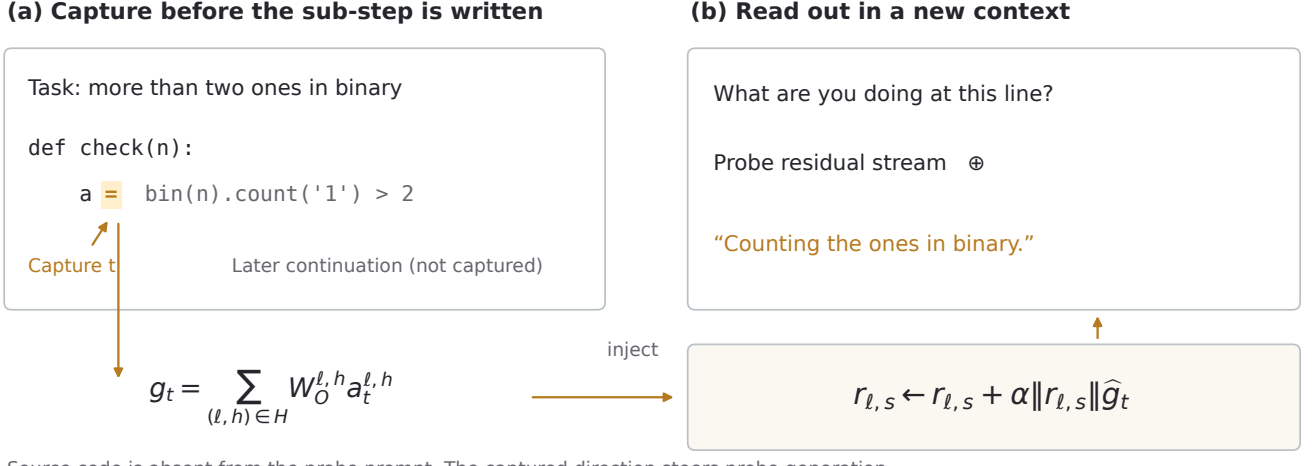


Shaded token: source capture ⊕: activation injection into a separate model run



Source code is absent from the probe prompt. The captured direction steers probe generation.

(a) Source computation

Check whether n has more than two ones in binary.

```
def check(n):
    a =   ← capture at t
```

$$g_t = \sum_{(\ell,h)\in H} W_O^{\ell,h} a_t^{\ell,h}$$

Selected head outputs → new context

(b) Describe

What are you doing at this line?

⊕ Assistant response

“Counting the ones in binary.”

(c) Implement

Check a condition in a C++ function.

```
bool check(int value) { return
```

⊕ `__builtin_popcount(value)>2;`